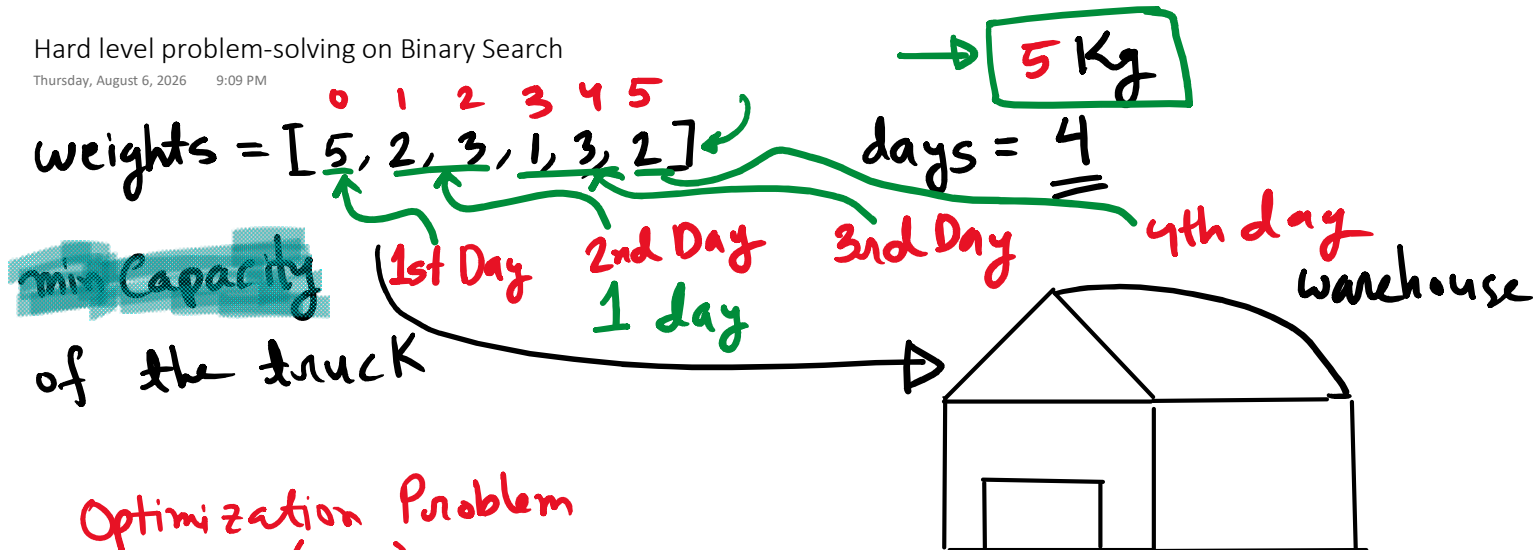
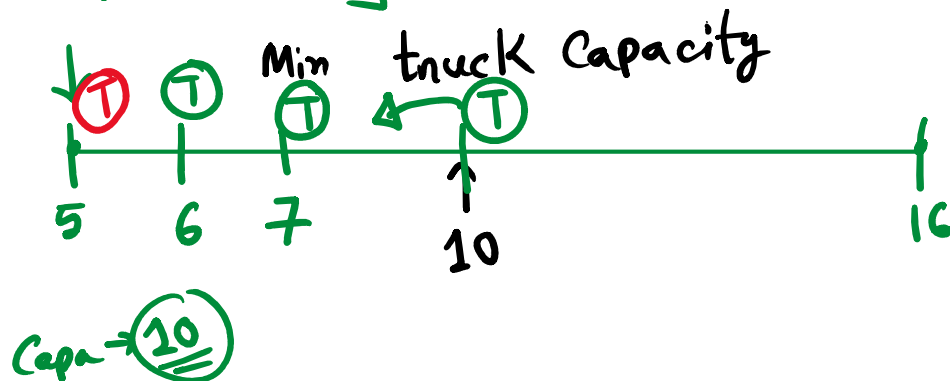
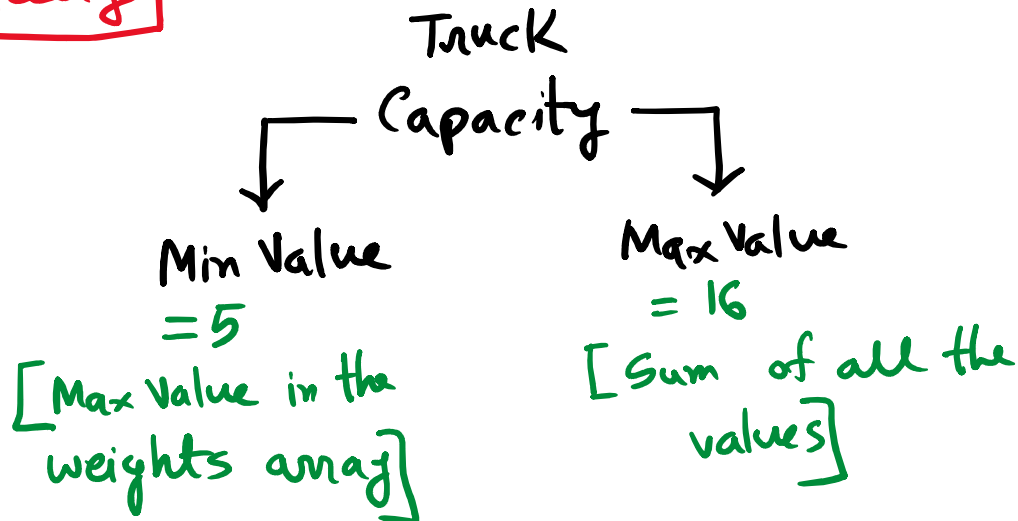




Optimization Problem

Greedy

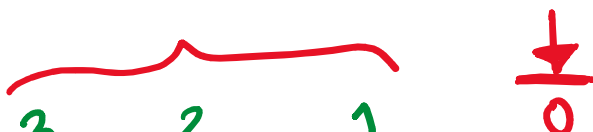
DP

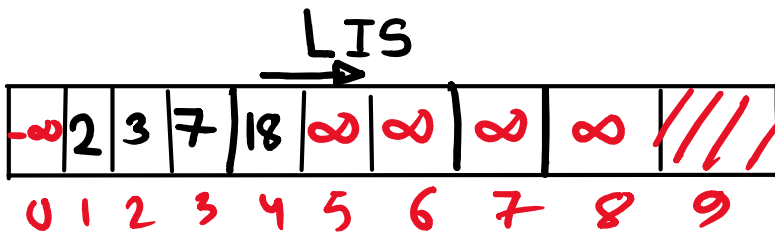
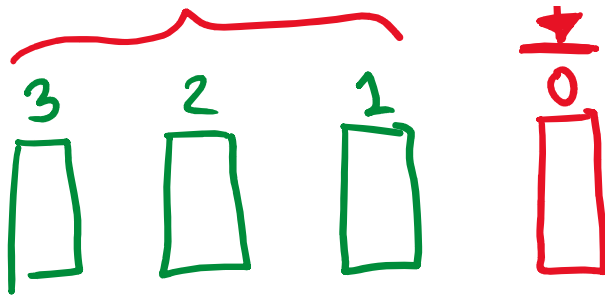


Input: weights = [1,2,3,4,5,6,7,8,9,10], days = 5
Output: 13

day = 7

From <<https://leetcode.com/problems/capacity-to-ship-packages-within-d-days/description/>>

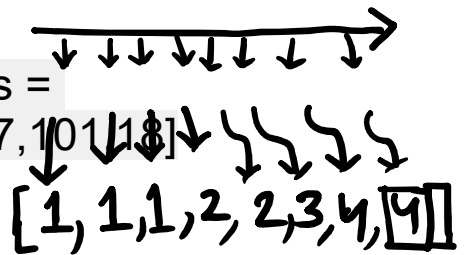




Input: nums =

[10, 9, 2, 5, 3, 7, 10, 1, 1, 8]

Output: 4



From

<https://leetcode.com/problems/longest-increasing-subsequence/>

2, 5, 7
2, 3, 7

[5, 10, 7] → [5, 7, 10]



Median

[5, 10, 7, 6] → [5, 6, 7, 10]

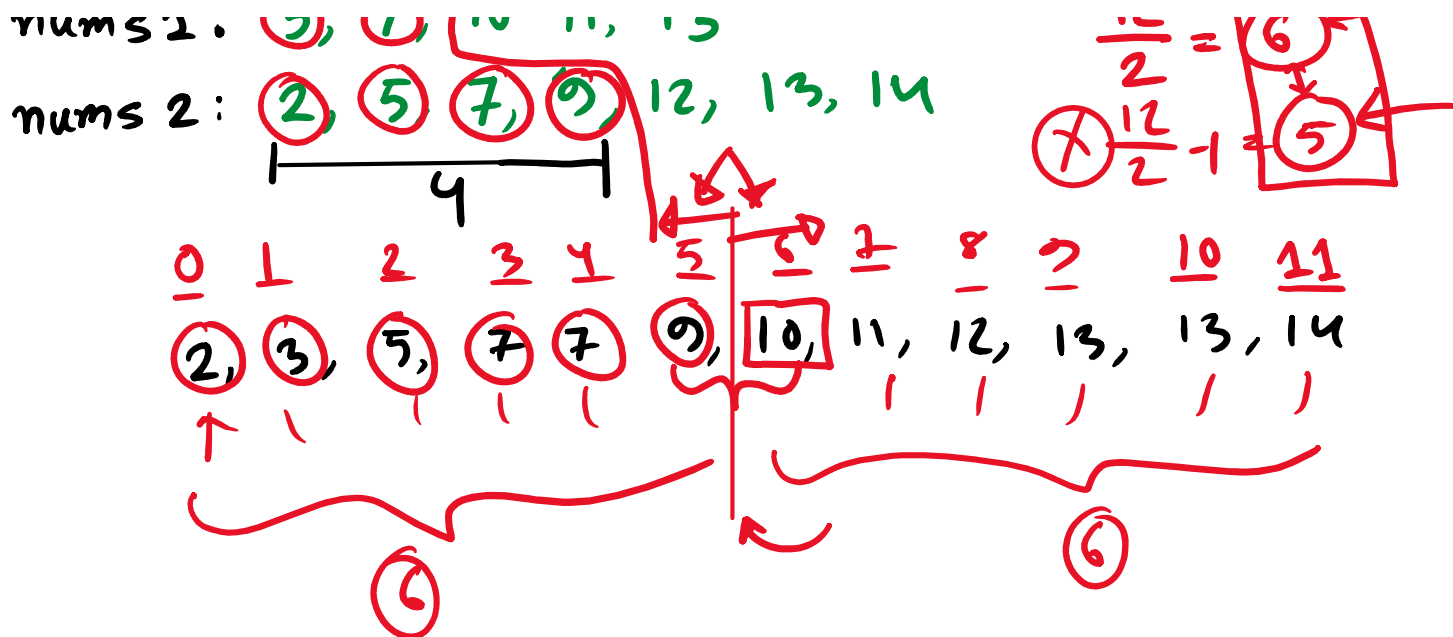
$$\frac{6+7}{2} = 6.5 \rightarrow \text{Median}$$

[0 - 5] → [3] → [1] → [2]

nums1: (3), (7), 10, 11, 13

nums2: (2), (5), (7), (9), 12, 13, 14

$$\frac{12}{2} = 6$$



<u>nums 1</u>	<u>nums 2</u>
0	6
1	5
2	4
3	3 ←
4	2
5	1
6	0 ← (X)